

TECHNICAL APPENDIX TO THE RELATIVISTIC SCALAR–VECTOR PLENUM

Field Equations, Lagrangian Formulation, and Bibliographic Commentary

Preface

This appendix provides the mathematical scaffolding of the *Relativistic Scalar–Vector Plenum* (RSVP) theory. It is not meant as proof but as a record of consistency: that the conceptual triad described in the *Island of Equations* and *Introduction to RSVP* can be rendered in formal language without recourse to speculation or metaphor. The presentation follows the custom of late-twentieth-century field theory, but its tone remains clerical rather than evangelical.

1. Field Definitions

Let $\Phi(x^\mu)$ denote a scalar field of informational capacity, $\mathbf{v}(x^\mu)$ a vector field representing local negentropic flow, and $S(x^\mu)$ an entropy density field. Together they form the RSVP triad:

$$\mathcal{R}(x^\mu) = \{ \Phi, \mathbf{v}, S \}.$$

We define the four-gradient $\nabla_\mu = \partial/\partial x^\mu$ and the spacetime metric $g_{\mu\nu}$ with signature $(+, -, -, -)$. The theory is formulated on a fixed but non-expanding manifold; gravitation emerges as a curvature of informational gradients rather than a dynamical geometry.

2. Lagrangian Density

The minimal Lagrangian capturing the RSVP dynamics is

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} (\nabla_\mu \Phi \nabla_\nu \Phi + \lambda \nabla_\mu S \nabla_\nu S) - \frac{1}{2} \alpha |\nabla_{[\mu} v_{\nu]}|^2 - \beta (\nabla_\mu v^\mu) S - V(\Phi, S), \quad (1)$$

where λ, α, β are coupling constants and $V(\Phi, S)$ encodes potential interaction terms such as

$$V(\Phi, S) = \frac{m_\Phi^2}{2} \Phi^2 + \frac{m_S^2}{2} S^2 + \gamma \Phi S.$$

The cross-term $\beta(\nabla_\mu v^\mu)S$ functions as an *entropic source*, allowing the divergence of the vector field to act as a local driver of entropy change.

2.1 Variational Equations

Variation of $\mathcal{L}$ with respect to $\Phi, \mathbf{v}$, and S yields:

$$\nabla_\mu \nabla^\mu \Phi + \frac{\partial V}{\partial \Phi} = 0, \quad (2)$$

$$\alpha \nabla_\nu \nabla^{[\nu} v^{\mu]} + \beta \nabla^\mu S = 0, \quad (3)$$

$$\lambda \nabla_\mu \nabla^\mu S - \beta \nabla_\mu v^\mu + \frac{\partial V}{\partial S} = 0. \quad (4)$$

Equations (2)–(4) form the canonical RSVP system. They describe coupled diffusive–advective behaviour between capacity, flow, and entropy. Solutions under stationary boundary conditions reproduce redshift-like attenuation and self-organizing relaxation without reference to cosmic expansion.

3. Conservation Laws

From Noether’s theorem applied to the symmetries of $\mathcal{L}$, one obtains conserved quantities:

- **Information Flux:** $J_\Phi^\mu = \nabla^\mu \Phi$ satisfies $\nabla_\mu J_\Phi^\mu = 0$ in the absence of potential coupling.
- **Entropy Continuity:** $\partial_t S + \nabla \cdot (\mathbf{v} S) = 0$ when $\beta = 0$; otherwise the divergence of $\mathbf{v}$ sources entropy production.
- **Energy Functional:**

$$E = \int d^3x \left[\frac{1}{2} (|\nabla \Phi|^2 + \lambda |\nabla S|^2) + \frac{\alpha}{2} |\nabla \times \mathbf{v}|^2 + V(\Phi, S) \right],$$

conserved under time translation symmetry.

These quantities provide the operational meaning of bookkeeping in RSVP: energy, entropy, and information are not distinct substances but conserved currencies within one ledger.

4. Entropic Interpretation

The coupling constant β in Eq. (??) measures the degree to which entropy production is sourced by vector divergence. For $\beta > 0$, regions of convergent flow ($\nabla_\mu v^\mu < 0$) act as local negentropic sinks, stabilizing structure. For $\beta < 0$, they generate entropic diffusion, mimicking redshift. This simple term replaces the need for an expanding metric: redshift becomes a scalar consequence of informational dilution.

5. Relation to Existing Theories

RSVP may be viewed as a generalization of several preceding frameworks:

- The **Einstein–Hilbert action** when Φ and S are constant, reducing to curvature dynamics.
- The **Klein–Gordon field** in the limit $\mathbf{v}=0, \beta=0$.
- The **Navier–Stokes entropy equation** when Φ is constant and $V=0$.

In this sense, RSVP is not a replacement for standard models but an interpolation among them, parameterized by informational curvature rather than spacetime expansion.

6. Linearized Solutions

For small perturbations around equilibrium ($\Phi_0, S_0, \mathbf{v} = 0$), linearization yields coupled wave–diffusion equations:

$$\partial_t^2 \delta\Phi - c_\Phi^2 \nabla^2 \delta\Phi = -\gamma \delta S, \quad (5)$$

$$\partial_t \delta S - D_S \nabla^2 \delta S = \beta \nabla \cdot \delta \mathbf{v}, \quad (6)$$

$$\partial_t \delta \mathbf{v} - \nu \nabla^2 \delta \mathbf{v} = -\beta \nabla \delta S. \quad (7)$$

Numerical integration of this system exhibits the characteristic RSVP behaviour: oscillatory diffusion, entropy-gradient lensing, and long-term relaxation toward finite coherence.

7. Annotated Bibliography

Thermodynamic Foundations

Boltzmann, L. (1872) *Weitere Studien über das Wärmegleichgewicht*. Introduced the statistical arrow of time as an emergent measure of disorder.

Gibbs, J. W. (1902) *Elementary Principles in Statistical Mechanics*. Formalized ensembles as descriptive rather than causal devices.

Landauer, R. (1961) “Irreversibility and Heat Generation in the Computing Process.” Demonstrated that logical erasure entails physical entropy production.

Informational and Entropic Gravity

Jacobson, T. (1995) “Thermodynamics of Spacetime.” Derived Einsteins equations from Clausius relation $\delta Q = T dS$.

Verlinde, E. (2011) “On the Origin of Gravity and the Laws of Newton.” Proposed gravity as entropic force; RSVP extends this without metric expansion.

Padmanabhan, T. (2015) “Gravity and Thermodynamics of Horizons.” Presented spacetime as an elastic medium obeying thermodynamic identities.

Information and Cognition

Bateson, G. (1972) *Steps to an Ecology of Mind*. Framed cognition as the perception of differences; a conceptual forerunner to informational field coupling.

Rosen, R. (1985) *Anticipatory Systems*. Linked biological order to internal models minimizing surprise—a cognitive analog to entropic descent.

Friston, K. (2009) “The Free-Energy Principle.” Proposed that self-organizing systems minimize an informational potential, mathematically akin to the RSVP Lagrangian.

8. Concluding Remarks

The equations presented herein are not revolutionary. They merely extend the bookkeeping of physics to include its own descriptive overhead. Entropy, information, and curvature are accounted as mutual liabilities rather than mysteries. The plenum does not expand; it reconciles. To write these equations is therefore to perform an act of balance: mathematics as the ethics of precision.

F.

Island of Equations, October 2025